

RoadEye

Gate 1 — Problem & Proof | Undergraduate

One-line problem summary

Motorcyclists, especially daily and long distance riders, can miss nearby hazards, blind spot situations, and crash consequences because current riding tools require attention away from the road and do not combine real time motorcycle sensing with hands free, eyes free safety alerts.

1. The Problem and Domain

Motorcyclists have less physical protection than occupants of cars and must continuously monitor traffic, road conditions and their own motorcycle. In practice, riders may rely on mirrors, phone navigation, separate Bluetooth devices and their own judgement. These approaches can leave blind spots and hazards unnoticed, while looking at or operating a phone can add distraction. A crash can also become more serious when the rider cannot quickly communicate their location to an emergency contact.

Domain: Transportation / Road Safety / IoT & AI-assisted mobility.

2. Who Exactly Has This Problem

Our initial target users are motorcycle riders in Sri Lanka who use their motorcycles regularly for commuting, delivery work, or longer distance travel, with an initial focus on riders who want navigation and safety assistance without repeatedly looking down at a phone.

The problem is intentionally narrower than “all road users”: RoadEye is designed around the specific combination of motorcycle blind spots, rider distraction, proximity hazards, crash detection and emergency communication.

3. Proof the Problem Is Real - Route B

We use Route B of the Gate 1 guide: a concise desk research validation, with the supporting source pack attached separately.

The problem is supported by the scale and safety context of motorcycle use. Sri Lanka's Department of Census and Statistics reports 4,856,979 registered motorcycles in 2023, making motorcycles a very large part of the country's vehicle population. WHO's Sri Lanka road safety profile reports 2,513 road traffic fatalities in 2021 and identifies powered two- and three-wheelers as a major vehicle group in the country. Globally, WHO states that motorcyclists and other powered two- and three-wheeled vehicle users account for nearly a quarter of annual road-traffic deaths, reinforcing that motorcycle safety is a meaningful road-safety problem rather than a niche convenience problem.

Our own project investigation also identified concrete rider pain points: limited awareness of surroundings, distraction from existing tools, helmet-related blind spots and delayed emergency response. These are the specific problems the RoadEye prototype is designed to address. What remains to be validated is the relative importance of each pain point for our chosen Sri Lankan rider

segments and their willingness to pay. We plan to validate those assumptions with rider interviews during the buildathon.

4. Market Size - TAM, SAM and SOM

We use the number of registered motorcycles as a transparent proxy for the potential user base. This is a unit-based market estimate, not a claim that every motorcycle owner will buy RoadEye. The Buildathon guide specifically asks us to show the assumptions behind each number.

Level	Calculation	Estimate	Basis
TAM	4,856,979 registered motorcycles × 100%	4,856,979 riders/vehicles	2023 Sri Lanka motorcycle registrations
SAM	4,856,979 × 25%	1,214,245	Assumption: first serviceable segment is 25% of motorcycles - regular urban/commuter, delivery and touring riders reachable by an initial Sri Lankan launch
SOM	1,214,245 × 0.5%	6,071	Assumption: first-year obtainable share is 0.5% of SAM; deliberately conservative for a new undergraduate-built product

These SAM and SOM percentages are our assumptions, not published market shares. They should be refined after user interviews, pricing tests and a clearer go-to-market plan. For a Gate 1 submission, the important point is that the assumptions are explicit rather than presented as unexplained market numbers.

5. Competitors and Alternatives

Alternative	What it provides	Gap / RoadEye angle
Garmin zūmo / zūmo R1 Radar	Motorcycle navigation, rider alerts, incident detection and, with the R1 Radar system, blind-spot/rear-approach alerts.	Strong navigation and rider-awareness products, but RoadEye aims to combine motorcycle sensing, helmet HUD/audio/haptics, crash detection and a mobile/backend system as one integrated safety experience.
Sena smart motorcycle products	Connected motorcycle helmets/intercom systems, audio, voice control and camera features.	Strong helmet connectivity, but RoadEye's differentiating focus is multi-sensor safety awareness and vehicle/helmet sensing rather than communication alone.
Phone navigation + Bluetooth headset	Maps, turn-by-turn navigation, calls/music and voice/audio directions.	Low switching cost and already familiar, but it does not inherently provide RoadEye's dedicated motorcycle proximity sensing, helmet HUD, crash sensing and integrated safety workflow.

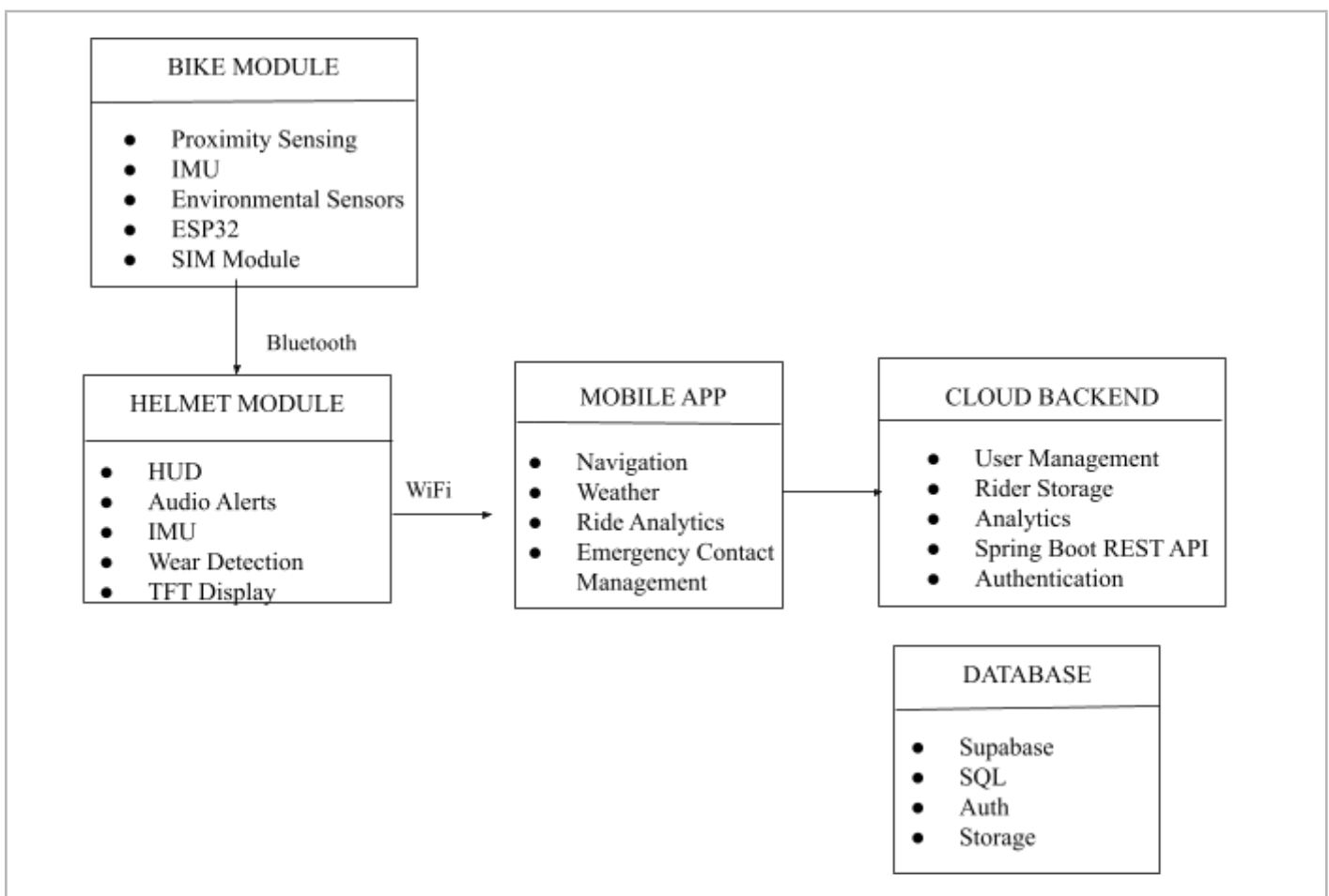
Mirrors + rider judgement	Traditional way to monitor rear/side traffic and road conditions.	Essential baseline, but depends heavily on continuous rider attention and cannot automatically combine sensor data or trigger emergency workflows.
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6. Our Solution

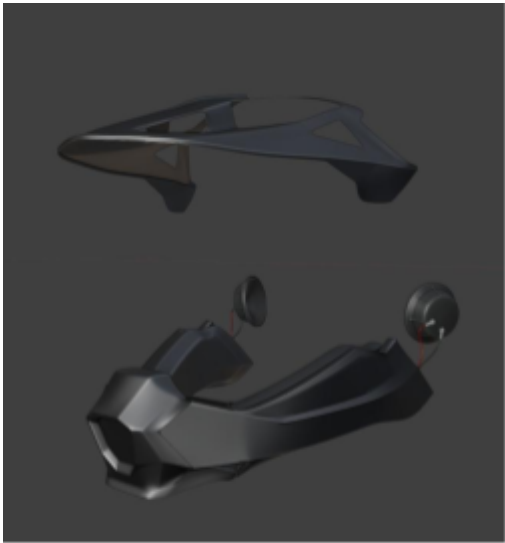
RoadEye is a smart motorcycle safety system built around a helmet module, bike module and mobile application. The bike module uses proximity sensing and an IMU to detect nearby vehicles and motorcycle motion, together with environmental sensing. The helmet module provides a heads-up display, audio alerts and haptic feedback so important information can reach the rider without requiring them to look down at a phone. The system also supports wear detection, navigation, ride analytics, crash detection and emergency-contact features.

The mobile application handles user authentication, navigation, preferences, emergency contacts and ride information. The backend uses a Spring Boot REST API and Supabase/PostgreSQL for data storage. The current project architecture already includes the major hardware, application, backend and deployment components; for the Buildathon, the focus is to turn the validated problem into a tighter AI-assisted safety prototype rather than simply reproducing every existing project feature.

Typical operating flow



Hardware Setup

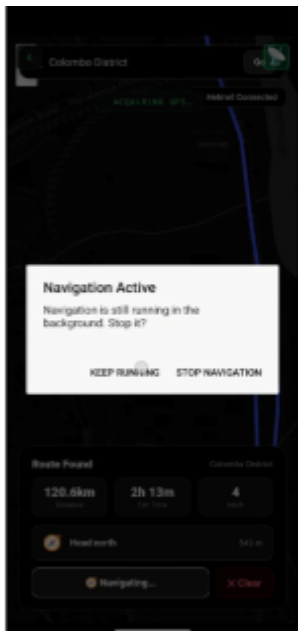
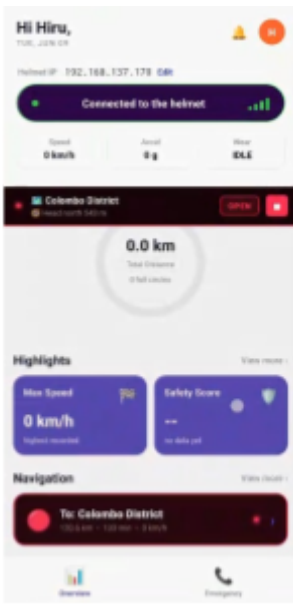
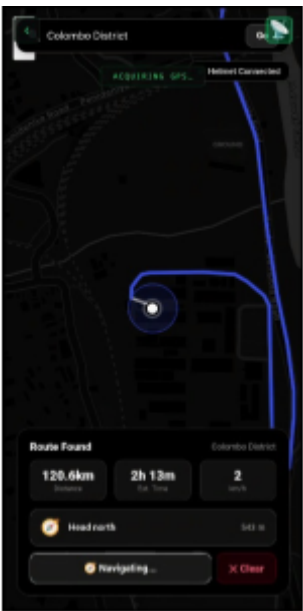
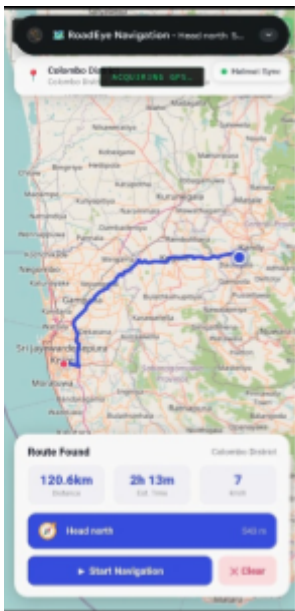


Helmet Module

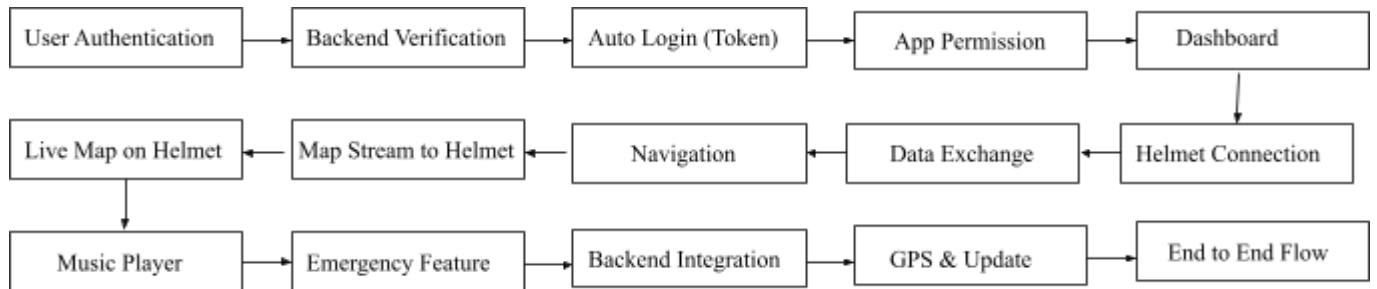


Bike Module

Mobile Application



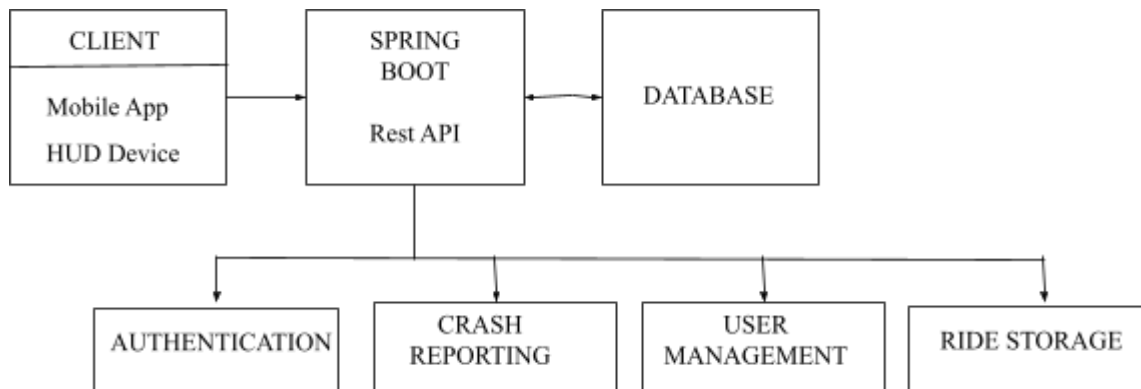
Mobile Application Flow



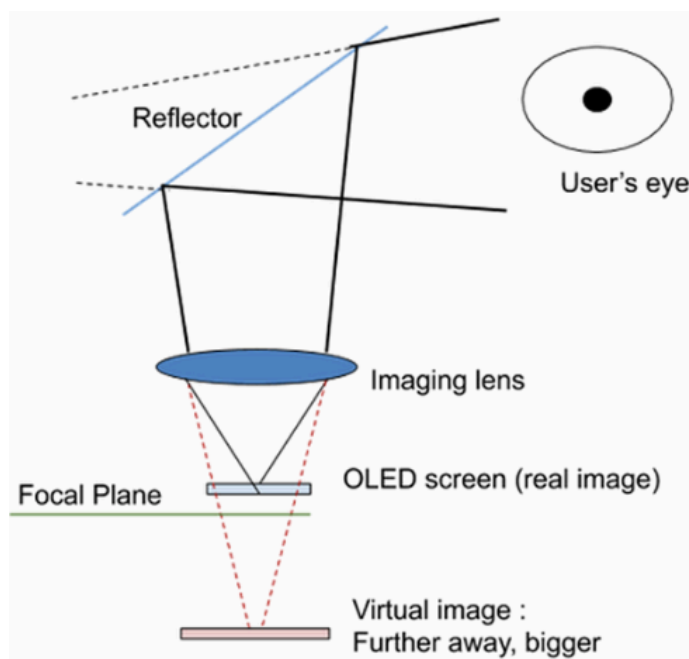
Helmet Hud design



Backend Architecture



How the Optic Works



Legality

Motor Traffic (Amendment) Act, No. 10 of 2019 - “152A. (1) No person shall use a telephone, mobile phone or any other movable communication equipment when driving a motor vehicle on a road.”

By law it is prohibited to operate any handheld device and give it attention while driving. Our design makes sure that answering phone calls does not require any hand operation and does not obstruct riders hearing like ear buds (speakers are embedded into the cushion)

Motor Traffic Act - “152. (1) No person shall drive a motor vehicle on a road when he is in such a position that he has not or cannot have full control of the vehicle or a full and unobstructed view of the road and the traffic ahead of the motor vehicle.”

The refractor of our heads up display is made of clear acrylic so that it does not cause any obstruction to the riders view. Our system does not require any hand inputs from the user so the rider always have full control of the vehicle

7. Why Does This Need AI?

AI is useful specifically for sensor-fusion and risk/event classification. Raw IMU, proximity and environmental signals can look different across riders, roads and normal riding events. A lightweight ML model can learn patterns that distinguish events such as a likely crash or unusually dangerous riding state from ordinary vibration, turning or braking. Deterministic thresholds remain as a safety fallback; AI is not being added just for branding. The three-week goal is a small, testable model rather than a claim of perfect autonomous crash prediction.

8. Technology Stack

Layer	Technology	Purpose
Embedded	ESP32, IMU, proximity sensors, environmental sensors, haptics, TFT/HUD, speakers	Real-time sensing and rider feedback
Mobile	Android/mobile app	Navigation, settings, emergency contacts, ride analytics
Backend	Spring Boot REST API, Java	Authentication, user management, ride/crash data and APIs
Database	Supabase PostgreSQL	Persistent relational storage
AI/ML	Python + lightweight ML model; model evaluation and inference integration	Sensor-event/risk classification
Deployment / tools	AWS Elastic Beanstalk, HTTPS/TLS, Git/GitHub, Postman	Deployment, security and testing

9. Three-Week Buildathon Plan

Week	Goal	Concrete outputs
Week 1 - Validate + data	Confirm the highest-priority rider pain points and define the AI task.	5+ rider conversations or a 25+ response survey; interview notes; competitor feedback; sensor-data collection protocol; cleaned event dataset; final MVP scope.
Week 2 - Build + integrate	Implement the minimum end-to-end safety loop.	Train/evaluate lightweight ML classifier; integrate inference with ESP32/mobile/backend path; implement HUD/audio/haptic alert flow; basic emergency workflow.
Week 3 - Test + demonstrate	Test realistic scenarios and package the evidence.	Confusion matrix/metrics; false-alert testing; crash/proximity scenario tests; usability feedback; bug fixes; deployed demo; final pitch and evidence pack.

10. Team and Scope Notes

RoadEye is being entered by an undergraduate team. The existing project work already covers helmet and bike hardware, mobile application, Spring Boot backend, Supabase/PostgreSQL storage, security and deployment.